

Zestril (Lisinopril): Ensuring Readiness for FDA Pre-Approval and Pre-License Inspections

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HCR 576: Drug Discovery, Development & Regulations

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November 27th, 2024

Pre-Approval Inspection Checklist for Zestril (Lisinopril)

1. Site Evaluation

- Ensures that manufacturing site is capable of manufacturing the drug safely.
- Staff should be properly trained.
- Staff and facility should have adequate equipment.

2. Data Integrity (Conformance to Application)

- Ensures that all data submitted to the regulatory authorities is accurate and truthful.
- Data that is submitted should be able to be authenticated by other information that has been collected by the manufacturer and pharmaceutical company.
- Conduct observations during manufacturing for comparison to the operations outlined in the application submitted to the regulatory authorities.

Pre-Approval Inspection Checklist

3. Readiness for Commercial Manufacturing: Key Steps

- Implement supplier qualification, lab readiness, and contamination control measures.
- Ensure a robust quality system and efficient batch release processes.
- Address discrepancies, investigations, and complaints effectively.

4. Data Integrity Readiness

- **Raw Data Review:** Ensure all raw data matches the application and is readily available to inspectors.
- **Traceability:** Test the traceability system to confirm all materials can be traced.
- **Data Security:** Verify the security of all data.
- **Record Accuracy:** Review records for correctness, completeness, and corrective actions (completed or needed).

5. Complete Development Report

- **Document Compilation:** Gather all relevant documentation and records for inspector review.
- **Structured Organization:** Arrange files to align with the submitted application.

Pre-License Inspection Checklist

1. Compilation of documentation and records

- Ensure copies of submitted documents are available for review.
- Records containing accurate data on batch information should include any deficiencies or issues that occurred during the manufacturing process.

2. Quality Control

- Investigator signatures should be present on all documents approving the facilities for production.
- Risk assessments should be conducted and documented to prevent deficiencies in products and during production.

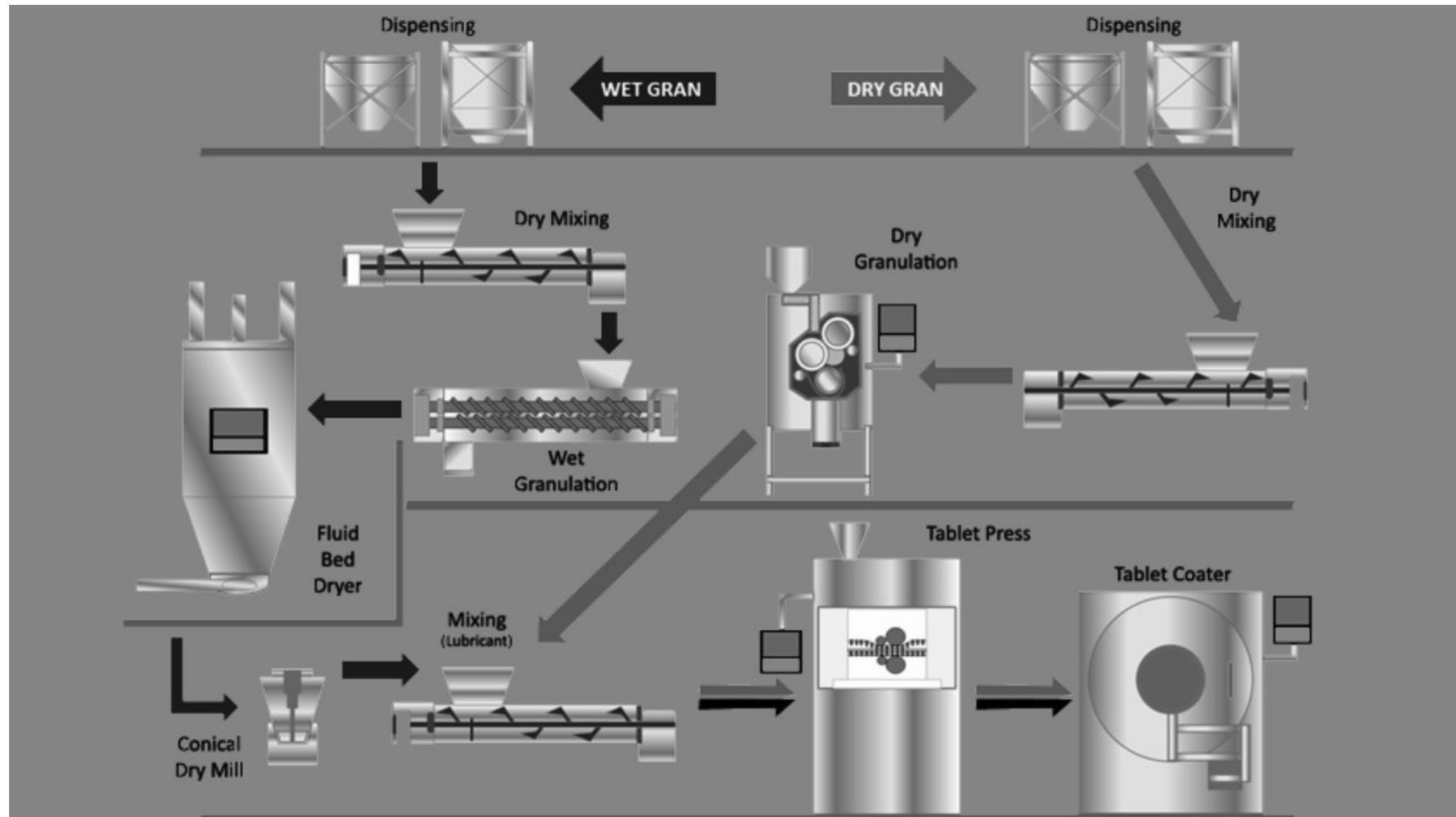
3. Inspection team

- Inspection team should be properly educated on identifying risks and deficiencies.
- Inspection team should produce an organized compilation of their findings, including corrective strategies or preventative measures.

Subject Matter Experts List

- **Regulatory Affair:** Ensures compliance with FDA regulations.
- **Quality Assurance:** Ensures quality systems and investigations.
- **Quality Control:** Reviews analytical data and testing methods.
- **Process Development:** Ensures process and scale-up drug manufacturing.
- **Manufacturing:** Oversees production process and controls.
- **Facilities:** Ensures maintenance and facility conditions.
- **Maintenance:** Work on equipments and ensures everything is working.
- **Product Manager:** Ensures day to day operations and manufacture.
- **Pharmacovigilance Management:** Ensures adverse event report and post-market safety.

Manufacturing

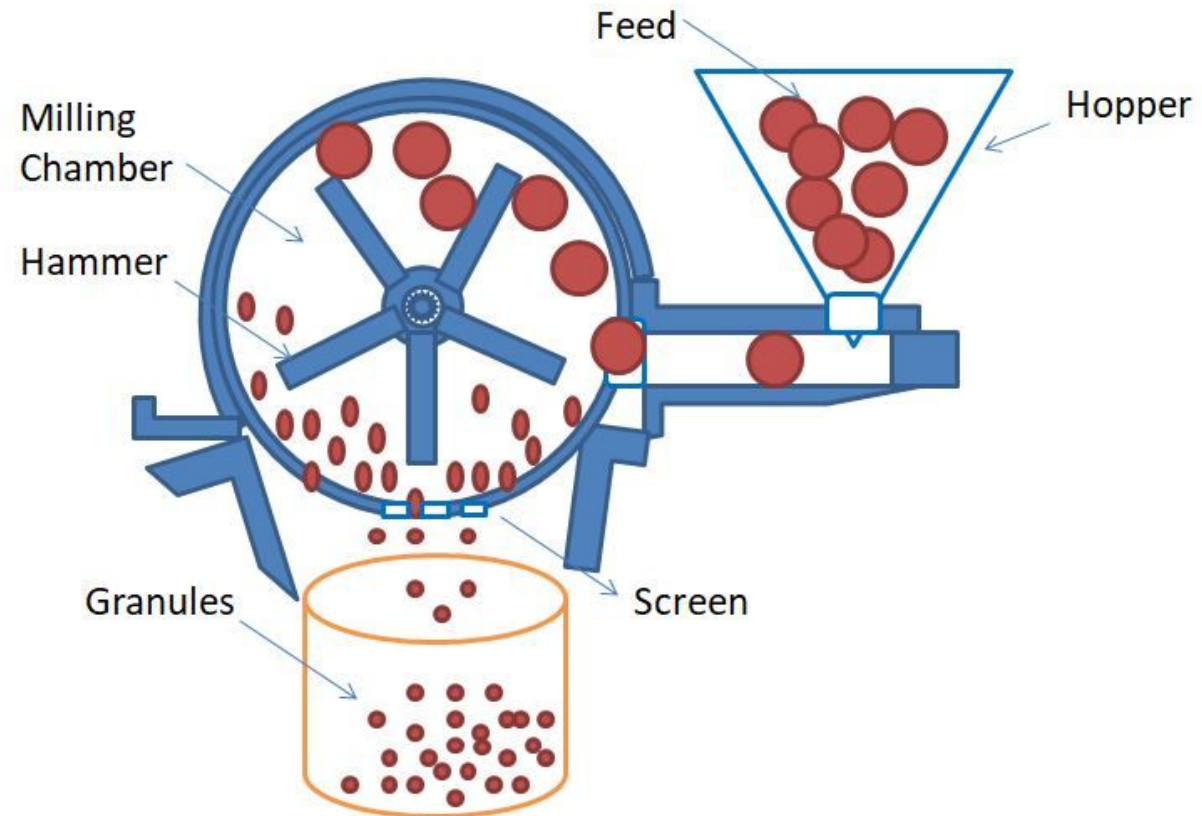


Manufacturing

- Milling & Screening:** Raw materials are milled and screened to achieve a uniform particle size.
- Blending:** The active pharmaceutical ingredient and excipients are mixed for a consistent distribution in each tablet.
- Granulation:** The powder blend is converted into granules to improve flowability and compressibility.
- Drying:** The granular material is dried to remove moisture and achieve the desired moisture content.
- Compression:** The lubricated granules are compressed into tablets using a rotary compression machine.
- Coating:** The tablets are film-coated.
- Inspection:** The tablets are inspected on an inspection belt for any out of specification tablets.
- Packaging:** The tablets are packaged in their specified packaging such as bottles, blister packs, etc.

Manufacturing

Milling and Screening



Blending



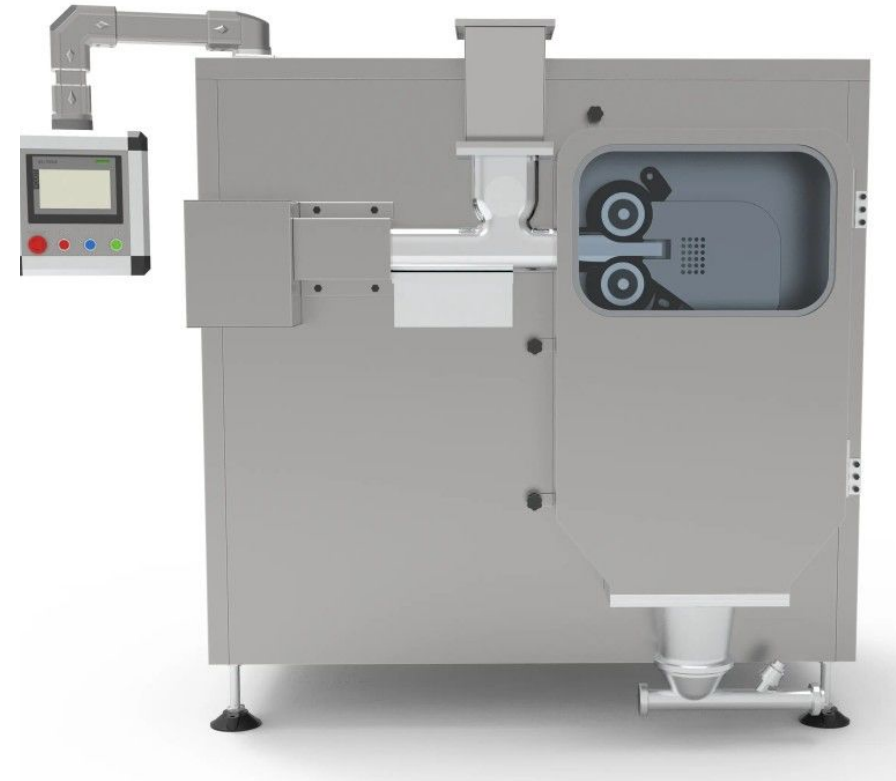
Manufacturing

Wet Granulation



Dryer (wet granulation only)

Dry Granulation



Manufacturing

Commercial Pill Press



Commercial Pill Coater



Manufacturing

Inspection



Packaging



Steps to Set Up an Inspection-Ready Room



Detailed Steps for Setting Up an Inspection Ready Room

Step 1: Develop a Standard Operating Procedure (SOP) for Managing Inspections

Objective: Establish clear guidelines for handling inspections.

Components: Include procedures for document handling, communication protocols, and the roles and responsibilities of team members. Helps ensure a consistent approach to inspections and reduces the risk of errors or miscommunication.

Step 2: Form an Inspection Readiness Team

Team Composition: Include quality managers, regulatory affairs specialists, subject matter experts (SMEs), and a dedicated note-taker.

Roles and Responsibilities: Assign each member specific tasks, such as leading facility tours, answering questions, and managing document flow.

Preparation: Conduct regular training and mock inspections to ensure the team is ready.

Detailed Steps for Setting Up an Inspection Ready Room

Step 3: Allocate Workspace for Inspectors and Audit Support

Inspection Room Setup: Ensure the room is clean, well-lit, and equipped with a large table, electrical outlets, Wi-Fi access, and a document management area.

Audit Support Room: Set up a nearby support room for the internal team to review and prepare documents before providing them to inspectors.

Communication Tools: Install a landline or use a reliable communication system for real-time updates.

Step 4: Pre-Inspection Walkthrough & Identify Subject Matter Experts (SMEs)

Facility Check: Perform a mock walkthrough to ensure all areas are organized, labeled correctly, and free from clutter.

SME Identification: Select experts who can confidently discuss technical aspects, such as manufacturing processes or quality controls, and answer inspector questions.

Detailed Steps for Setting Up an Inspection Ready Room

Step 5: Audit Records that FDA is Likely to Review

Document Review: Identify and organize critical records, including SOPs, training logs, batch records, and complaint logs.

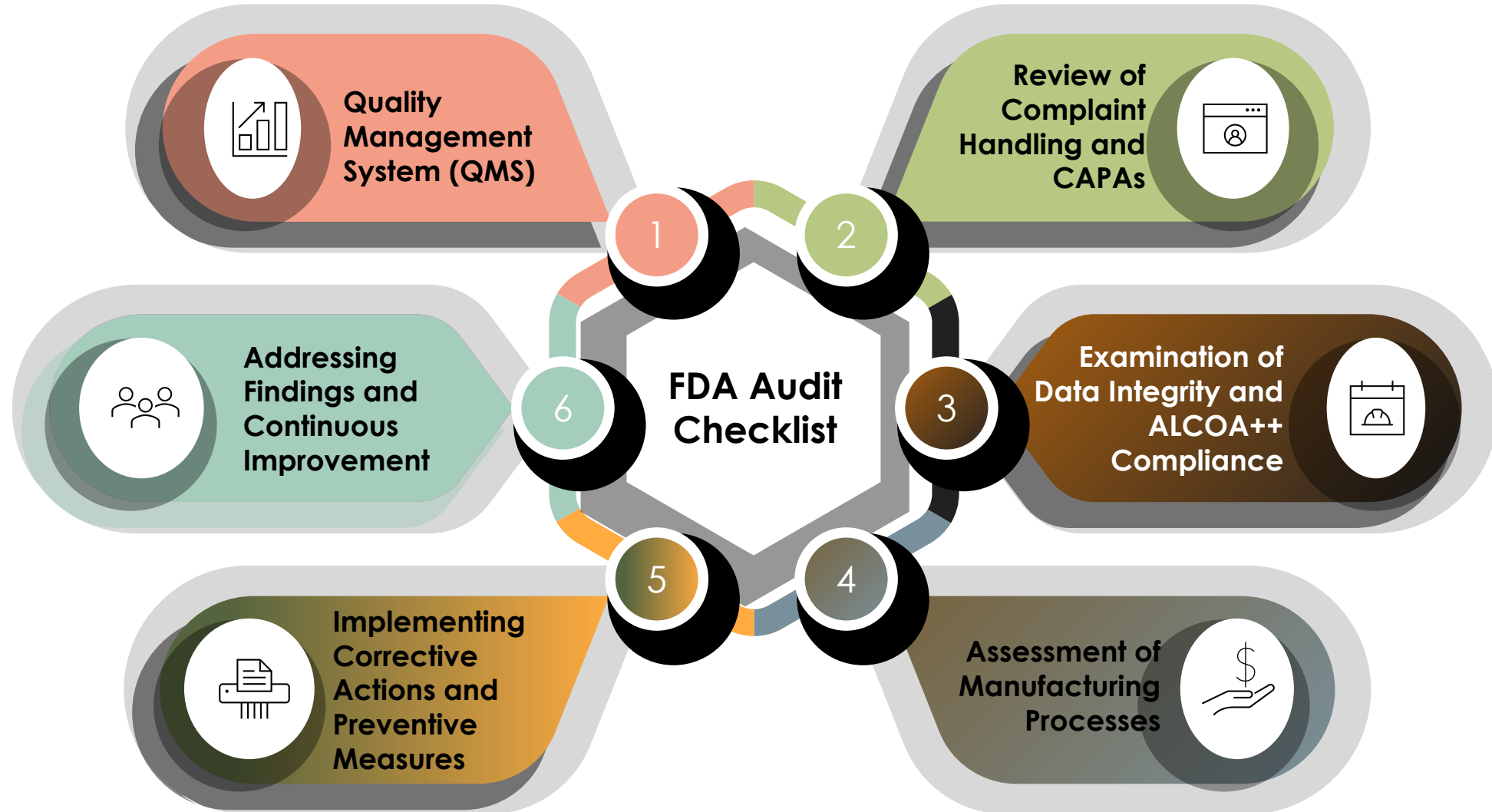
Pre-Audit Checks: Ensure all documents are complete, signed, and up to date. Address any discrepancies before the inspection.

Step 6: Create a Justification Strategy for Key Decisions and Closing Meeting Preparation

Decision Justification: Develop evidence-based explanations for key decisions made during product development and manufacturing.

Practice Responses: Train staff to give concise, accurate answers and to avoid volunteering unnecessary information. Prepare for the final meeting with inspectors to discuss observations and provide clarifications. Only key staff members should attend.

Key Topics for FDA Review During Inspection



Key Topics for FDA Review During Inspection

1. Quality Management System (QMS) Evaluation

The FDA evaluates the Quality Management System (QMS) by analyzing compliance with regulatory standards, focusing on SOP effectiveness, personnel training, change control, and documentation practices to ensure product quality and efficiency.

2. Complaint Handling and Corrective and Preventive Actions (CAPAs) Review

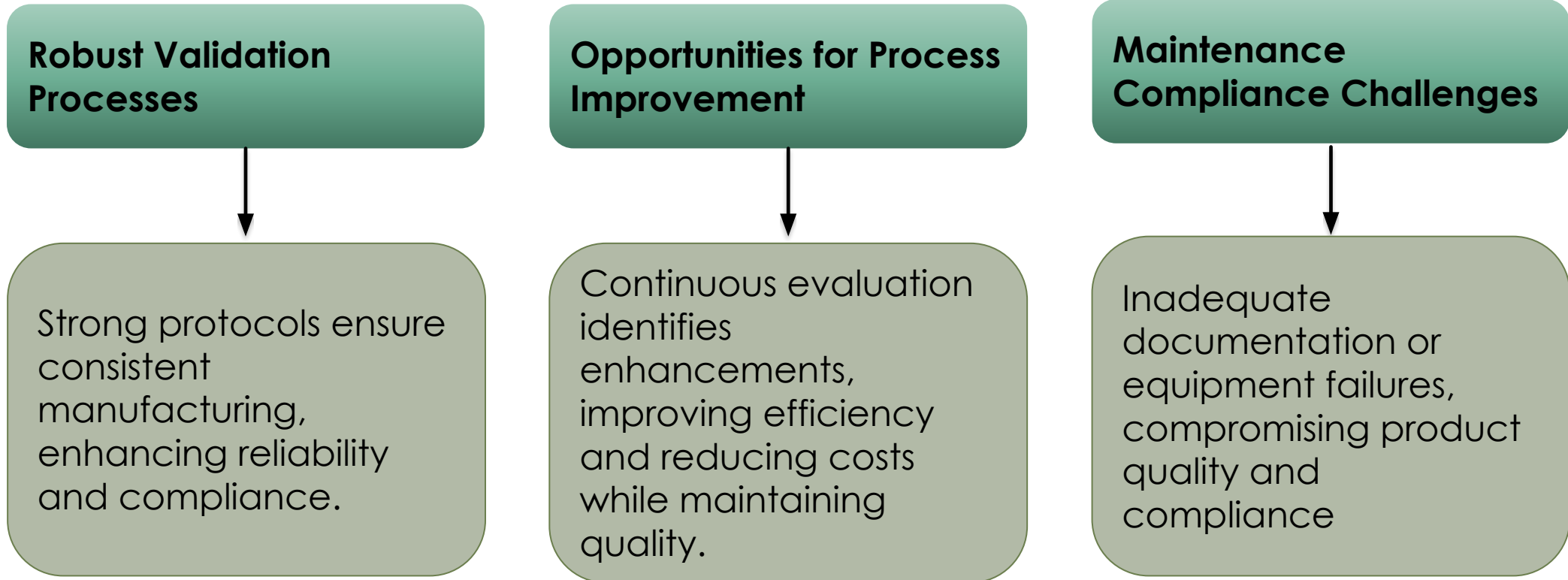
Thorough root cause analysis is crucial for effective CAPA, as it addresses complaints, identifies systemic issues, and enhances product quality and compliance.

3. Data Integrity and ALCOA++ Compliance

- **Data Traceability:** Ensure entries are **Attributable** to specific individuals, enhancing accountability.
- **Real-Time Recording:** Record data **Contemporaneously** to reduce errors and support audits.
- **ALCOA++ Principles:** Maintain data that is Attributable, Legible, Contemporaneous, Original, Accurate, Complete, Consistent, Enduring, and Available.

Key Topics for FDA Review During Inspection

4. Assessment of Manufacturing Processes



Key Topics for FDA Review During Inspection

5. Implementing Corrective Actions and Preventive Measures

Systematic CAPA Framework: A structured CAPA system addresses non-conformities through thorough documentation, prompt corrective actions, and continuous monitoring to maintain compliance and improve product quality.

Supplier Risk Assessment: A robust risk assessment framework identifies and mitigates supply chain vulnerabilities, ensuring material quality and compliance, which bolsters product safety and efficacy.

6. Addressing Findings and Continuous Improvement

Handling Deviations and Investigations: A structured approach ensures timely reporting, thorough investigations, and effective CAPA to maintain compliance and product quality.

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